

Wicaksono Leksono Muhamad

AI Engineer | Machine Learning Engineer | NLP Engineer | AI Researcher | NLP Researcher | Data Scientist
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SUMMARY

AI/ML engineer and researcher with 7 years of experience building customer-facing AI solutions spanning agentic AI workflows, retrieval-augmented generation (RAG) systems, and production LLM deployments on Google Cloud Platform. Published first-author at ACL venues with expertise in LangGraph, LangChain, and Model Context Protocol (MCP) for multi-agent orchestration. Delivered cross-university and cross-country collaborative platforms serving real end-users across healthcare, education, and climate policy domains.

EDUCATION

Universitas Gadjah Mada <i>Master of Artificial Intelligence</i>	Yogyakarta, Indonesia Feb. 2024 – Oct. 2025
Institut Teknologi Sepuluh Nopember <i>B.A.Sc. Computational and Applied Mathematics</i>	Surabaya, Indonesia Aug. 2018 – Nov. 2022

EXPERIENCE

Research Assistant, Machine Learning Systems <i>Universitas Gadjah Mada</i> Proposed early-exit training and inference method for energy-efficient and low-latency model deployment on resource-constrained hardware.	Feb. 2026 – Present Yogyakarta, Indonesia
Research Assistant, Software Engineering Lab <i>Universitas Gadjah Mada</i> - Built LLM-based NLP tooling for requirements engineering; added pragmatic and semantic testing on user story quality specifications, extending prior automated QUS analysis. - Led machine learning workstream for national grant; built PHQ-aligned LLM screening with deep learning facial affect analysis, evaluated on 302 samples. Presented results to university stakeholders and grant reviewers.	Mar. 2025 – Oct. 2025 Yogyakarta, Indonesia
Researcher, Natural Language Processing <i>Mantera Studio Research Lab</i> - Published first-author paper at SemEval-2026 Task 11; achieved top-5 of 1200+ submissions on multilingual syllogistic reasoning with 93% accuracy via prompt engineering and deterministic parsing. - Co-authored BEA 2026 Shared Task on multilingual lexical complexity prediction; built transformer ensemble with contrastive learning and regression objectives across Spanish, German, and Mandarin Chinese.	Feb. 2022 – Present Jakarta, Indonesia
Web Developer <i>Mastej Studio</i> Built React/Vite tourism platform (400+ DAU, 100 leads/day) and apartment marketing site with 360-degree virtual tour (10 daily inquiries); delivered demos and prototypes directly to regional business clients.	2019 – 2022 Malang, Indonesia

PUBLICATIONS

Aligning Implied Statements for Implicit Hate Speech Generalizability <i>First Author – Under Peer Review: ACL Rolling Review (OpenReview)</i> - Trained triplet contrastive learning objective with context-bounded semi-hard negatives to separate near-surface pairs in hate speech text embeddings. - Improved out-of-domain F1-score by 4–8% across 3 hate speech NLP benchmarks over contrastive learning baselines.	Jan. 2026
Improving Lexical Difficulty Prediction with Context-Aligned Contrastive Learning <i>First Author – ACL-BEA 2026 Shared Task (arXiv:2605.08950)</i> Proposed Context-Aligned Contrastive Regression: Ridge ensemble over fine-tuned transformers (XLM-RoBERTa, mDeBERTa) with Cross-View Context and Ordinal Soft Contrastive Learning losses.	May 2026

- Achieved top-15 across all L1 groups on BEA 2026 benchmark; ranked 14th (Spanish), 11th (German), 7th (Mandarin Chinese).

Normalization and Deterministic Parsing for Formal Reasoning in LLMs

Mar. 2026

First Author – ACL-SemEval-2026 (arXiv:2603.02676)

- Introduced structural abstraction transforming syllogisms into canonical logical representations with deterministic parsing to reduce content effects in multilingual LLM reasoning.
- Top-5 submissions across all subtasks, outperforming fine-tuning and activation-level interventions.

Automated Detection of Quality Issues in User Stories – Co-Author, ICODSE 2025

2025

On-Edge Device Optimization for Audio Classification – First Author, IEEE Xplore

2025

- Deployed MFCC-based deep learning classifier on Arduino Nano 33 BLE; 95.2% accuracy, 3.00 ms inference latency, 3.80 KB memory footprint via model optimization.

PROJECTS

Dyslexic Reader | Dart, Next.js, Docker, Kubernetes, LLM | dyslexic.app

2026

- Built LLM-powered dyslexia support application in collaboration with Queen's University Belfast x UGM; deployed on Kubernetes with Dart mobile and Next.js web frontends.
- Implemented dyslexia lens (real-time font rendering transform), LLM summarization, text professionalization, and definition features; integrated AHRQ screening protocol for clinical alignment.

Mental Health Screening Platform | Flask, LangGraph, LangChain, PyTorch | mentalhealth.laptopmerahputih.id Nov. 2025

- Shipped LLM-driven PHQ + CPAS screening workflows with agentic AI orchestration via LangGraph; served 40+ concurrent user sessions with stable uptime.
- Integrated facial affect capture via deep learning; reduced p95 latency and cost by 90% through local response batching and request queuing. Monitored with LangSmith telemetry.

IndoClimate Chatbot | React, LangGraph, LangChain, RAG, OpenAI API | chat.indoclimate.id

Aug. 2025

- Developed WhatsApp webhook + REST API with retrieval-augmented generation over Indonesian climate legislation for UGM x Deakin University; presented prototype demos to cross-university technical stakeholders.
- Built LLM summarization pipeline for long legal documents to improve vector similarity matching in ChromaDB; monitored agent traces and query performance via LangSmith.

TASYA Academic Chatbot | Flask, LangGraph, LangChain, ChromaDB, OpenAI | tasya.dtmi.ugm.ac.id Feb. 2025

- Built WhatsApp webhook + REST API for academic support RAG chatbot; indexed ~2,000 pages across 7 institutional documents with chunking pipeline into ChromaDB vector database.
- Orchestrated multi-step retrieval and response generation via LangGraph agentic workflows; tracked query traces and retrieval quality with LangSmith observability.

SOC Dashboard | Vite.js, Redux Toolkit, Go, Langsmith, Langgraph

Aug. 2024

- Refactored frontend into domain-driven design; reduced redundant code by 70%. Integrated server telemetry ingestion, distributed tracing, audit logging, and LLM-based log summarization.

Sijinak Attendance System | gRPC, FFI, Dart, Python | GitHub

2025

- Built RFID-based attendance system integrating Hikvision hardware via Foreign Function Interface to native C SDK; exposed microservice layer over gRPC with Dart frontend.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript/JavaScript, Go, Dart, Java, C/C++, Rust, SQL (PostgreSQL)

Machine Learning & AI: PyTorch, TensorFlow, HuggingFace Transformers, scikit-learn, deep learning, NLP, LLM, fine-tuning, model evaluation, embeddings, contrastive learning, knowledge distillation, model optimization

LLM & Agentic AI: LangGraph, LangChain, LangSmith, Model Context Protocol (MCP), retrieval-augmented generation (RAG), prompt engineering, OpenAI API, vector databases (ChromaDB), agentic workflows, multilingual NLP

Web & Backend: React, Next.js, Flask, Node.js, Vite.js, REST API, gRPC, API gateways, microservices

Infrastructure & Observability: Docker, Kubernetes, Git, Linux, nginx, Google Cloud Platform, CI/CD, distributed tracing, logging, audit logging, $\LaTeX$

Data Science: Pandas, NumPy, Matplotlib, data analysis, data visualization, experiment tracking

Languages: Bahasa Indonesia (Native), English (Professional – TOEFL ITP 587), German (B1), Javanese (Conversational)